

$$u := \exp\left(\sum_{k=2}^{49} \int_0^{\frac{1}{k}} \prod_{n=1}^{\infty} (1+x^{2^n}) dx\right)$$

$$P_N := \prod_{n=1}^N (1+x^{2^n})$$

$$P := \lim_{N \rightarrow \infty} \prod_{n=1}^{\infty} (1+x^{2^n})$$

$$P_N = (1+x^2)(1+x^4)\dots(1+x^{2^N})$$

$$(1-x^2)P_N = (1-x^2)(1+x^2)(1+x^4)\dots(1+x^{2^N})$$

$$(1-x^2)P_N = (1-x^4)(1+x^4)(1+x^8)\dots(1+x^{2^N})$$

$$(1-x^2)P_N = 1-x^{2^{N+1}}$$

$$P_N = \frac{1-x^{2^{N+1}}}{1-x^2}$$

$$P = \lim_{N \rightarrow \infty} \frac{1-x^{2^{N+1}}}{1-x^2}$$

$$\text{If } |x| < 1 \implies \lim_{N \rightarrow \infty} x^{2^{N+1}} \rightarrow 0$$

$$\therefore P = \frac{1}{1-x^2} \text{ s.t } |x| < 1$$

$$u = \exp\left(\sum_{k=2}^{49} \int_0^{\frac{1}{k}} \frac{1}{1-x^2} dx \text{ s.t } |x| < 1\right)$$

$$\exp\left(-\frac{1}{2} \sum_{k=2}^{49} \ln\left|\frac{x-1}{x+1}\right| \Big|_0^{\frac{1}{k}} \text{ s.t } |x| < 1\right)$$

$$\exp\left(-\frac{1}{2} \sum_{k=2}^{49} \ln\left(\frac{1-k}{1+k}\right)\right)$$

$$\exp\left(\frac{1}{2} \sum_{k=2}^{49} \ln\left(\frac{k+1}{k-1}\right)\right)$$

$$\exp\left(\frac{1}{2} \ln \prod_{k=2}^{49} \frac{k+1}{k-1}\right)$$

$$\exp\left(\frac{1}{2} \ln \prod_{k=2}^{49} \frac{k+1}{k-1}\right)$$

$$\exp\left(\frac{1}{2} \ln \frac{3 \cdot 4 \cdot 5 \dots 50}{1 \cdot 2 \cdot 3 \dots 48}\right)$$

$$\exp\left(\frac{1}{2} \ln \frac{\cancel{3} \cdot \cancel{4} \cdot \cancel{5} \dots 50}{1 \cdot 2 \cdot \cancel{3} \cdot \cancel{4} \cdot \cancel{5} \dots \cancel{48}}\right)$$

$$\exp\left(\frac{1}{2} \ln \frac{49 \cdot 50}{2}\right)$$

$$\exp\left(\frac{1}{2} \ln \sqrt{1225}\right)$$